End-to-End Supply Chain Analytics, Demand Forecasting & Delivery Delay Prediction System

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Abstract

This project presents an End-to-End Supply Chain Analytics and Delivery Delay Prediction System integrating Data Analysis, SQL Business Intelligence, Demand Forecasting, Machine Learning, and Power BI visualization. The solution analyzes more than 16,000 orders and evaluates sales, profit, inventory, demand patterns, and delivery performance.

ATS-Friendly Project Summary

Developed an End-to-End Supply Chain Analytics, Demand Forecasting, and Delivery Delay Prediction System using Python, SQL, XGBoost, and Power BI. Analyzed 16K+ orders, ₹3.1M+ sales, and ₹346K+ profit to generate business insights, forecast 53K+ demand, identify inventory risks, and predict delivery delays.

Business Problem Statement

Modern supply chains face delivery delays, demand uncertainty, inventory shortages, excess inventory costs, supplier performance variability, and inefficient logistics planning. The objective is to build a data-driven analytical framework for monitoring performance, predicting delays, forecasting demand, and optimizing inventory.

Dataset Overview

The dataset contains operational, logistics, customer, financial, and product information including order dates, delivery dates, sales, profit, discounts, shipping modes, markets, regions, and product categories.

Data Cleaning & Preprocessing

Performed missing value handling, date standardization, feature engineering, label encoding, and business metric creation including month, quarter, weekday, profit margin, discount amount, and weekend indicators.

Exploratory Data Analysis

Analyzed sales performance, profit trends, regional performance, delivery delays, inventory patterns, and customer demand trends to identify operational bottlenecks and opportunities.

SQL Analysis

Used SQL to identify top-performing markets, profitable product categories, regional sales performance, customer demand trends, and delivery delay patterns.

Demand Forecasting Methodology

Historical demand trends were analyzed to forecast future inventory requirements, identify seasonal patterns, and generate reorder recommendations for inventory planning.

Delivery Delay Prediction Model

Developed an XGBoost multi-class classification model to predict Early, On-Time, and Late deliveries using operational, financial, and logistics features. Model accuracy was approximately 60% with average prediction confidence of 67.9%.

Business Insights

Analyzed 16K+ orders, ₹3.1M+ sales, and ₹346K+ profit. Europe emerged as the highest revenue-generating market. Delivery delay rate reached 57.7%, highlighting logistics optimization opportunities.

Prediction Insights

Late deliveries represented the largest shipment category. High-discount orders exhibited elevated delivery risk. Predictive analytics enabled early identification of high-risk shipments.

Business Impact & ROI

Forecasted 53K+ future demand, generated 10K+ reorder recommendations, improved supply chain visibility, supported inventory optimization, and enabled proactive logistics planning.

Technology Stack

Python, Pandas, NumPy, Scikit-Learn, XGBoost, SQL, Power BI, Machine Learning, Predictive Analytics, GitHub.

Future Scope

Future enhancements include cloud deployment, MLOps integration, real-time monitoring, automated alerts, supplier risk prediction, and advanced forecasting models.

Conclusion

The project successfully integrates analytics, forecasting, machine learning, SQL, and business intelligence to improve supply chain efficiency, reduce operational risks, and support data-driven decision-making.

GitHub Repository Link

https://github.com/shubh97-hub/End-to-End-Supply-Chain-Analytics-Delivery-Delay-Prediction-System

Dashboard